PD20 open questions — K draws and ρ^* (explainer)

Last synced: 2026-08-13
Audience: Charles — print, edit, digest before next Alex touchpoint
Context: PD19 (Gibbs SELECT + MLE ambition); PD20 (t in denominator; inverted-U sweep before MLE)
Related: `../transcripts/PD20_notes.md` · `../transcripts/PD19_notes.md` · `MLE_prob.docx` · `PD19_board.jpeg`
Status: Both questions are **still open**. PD20 locked temperature; neither K-draw semantics nor ρ^* estimation path is settled.

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1. The K-draws problem

What the formula on the board says

$$P(Y_i = 1) \;=\; \frac{\exp(S_i/t)}{\sum_{k=1}^{N} \exp(S_k/t)}$$

This is a **softmax** over the whole season pool. Critical identity:

$$\sum_{i=1}^{N} P(Y_i = 1) \;=\; 1$$

If you treat each $P(Y_i = 1)$ as a **literal Bernoulli probability** for player i, the model says: **expected number drafted ≈ 1 per season**, not **K ≈ 60** (or ~1% of N).

That is the tension with MBB: many player-seasons, **few** draft slots.

Three objects people often merge

Object	What it is	Sum over pool
Gibbs weight	$w_i \propto \exp(S_i/t)$	weights sum to 1 after normalize
Bernoulli p_i	"each i drafted independently with prob p_i"	$\sum p_i$ can be anything if you design it
Exactly K winners	NBA draft: K names per season	always **K**, not 1

PD19's written formula looks like **Bernoulli p_i = softmax**. PD20's "equivalent to top-K at cold t" sounds like **exactly K winners**. Those are **not the same** unless you add an extra step.

What SELECT rule `"D"` does (planned bridge)

1. Compute **scores** S_i (SCORE — unchanged).
2. Gibbs **weights** $w_i \propto \exp(S_i/t)$.
3. **Draw K players without replacement** using those weights (same skeleton as old rule `"A"` in code).

That gives **exactly K** `Y_selected = 1` per season for **sim plots** — same count as PD17 top-K.

But then $P(Y_i = 1)$ **for MLE is not** the simple softmax formula above, unless you define:

- $P(Y_i = 1)$ = **marginal** probability that i is in the K-set under K draws without replacement (Plackett-Luce), or
- you use softmax only as **weights** for drawing, not as the literal likelihood term.

Why it matters for the inverted-U sweep (PD20 first job)

For the **immediate task** (sweep t, replot Hero bins):

- **K draws without replacement** is the right sim readout — comparable to old top-K, same K per season, same binning.
- Alex's worry ("does the U survive softening the cut?") is about **replacing the hard rank cut with a stochastic rule that still picks K people** — not about fitting a literal softmax Bernoulli yet.

No blocker for the t sweep if we use rule `"D"` = Gibbs weights + **K draws**.

Why it matters later for MLE

When you write **likelihood** on real $Y_i \in \{0, 1\}$:

Approach	Likelihood	Issue
Independent Bernoulli with p_i = softmax	$\prod_i p_i^{Y_i}(1-p_i)^{1-Y_i}$	$\sum p_i = 1$ → ~1 draft/season; **wrong scale** for MBB
Bernoulli with $p_i = K \cdot \mathrm{softmax}_i$ (capped at 1)	ad hoc	$\sum p_i \approx K$ but not standard; correlation ignored
One K-set per season (Plackett-Luce)	proper for "K without replacement"	harder; not the single-line softmax
Logistic on S_i: $P(Y_i = 1) = \sigma(\alpha + \beta S_i)$	classic rare-event	**no** "sum to 1" constraint; not full Gibbs pool

Open question for Alex: For MLE, is the object **(a)** Gibbs weights + K draws (generative sim), or **(b)** literal $P(Y_i = 1)$ = softmax (the written formula)? PD20 prioritized **(a)** for plots; PD19 synopsis read like **(b)**.
One sentence: Softmax is natural for **relative** chance among players; it is **not** automatically the Bernoulli p_i for **K simultaneous** winners unless you define the multi-draw process.

Cold / hot t (with t in the denominator)

t	Effect
Small t (cold)	Weights ≈ point mass on high S_i → K draws ≈ **top-K** (PD17 world)
Large t (hot)	Weights ≈ uniform → K draws ≈ **random K** → inverted-U may flatten (Alex's fear)

That is what the t sweep is meant to test.

Glossary — softmax (same math as Gibbs here)

Softmax turns scores S_i into **probabilities that sum to 1**:

$$P_i = \frac{\exp(S_i/t)}{\sum_j \exp(S_j/t)}$$

Higher S_i → higher P_i. Smaller t → sharper (more winner-take-all). Larger t → flatter (more equal).

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2. The ρ^* question

What ρ actually does

ρ **lives only in ASSIGN** (LG):

$$\tilde{w}_{ij} = R_j \exp(-\rho|\hat{A}_i - \hat{\mu}_j|)$$

It controls **who sits with whom** → rosters → team L_C → then S_i → then SELECT.

ρ **does not appear in** $S_i = \hat{A}_i - \lambda L_C$.

What the PD19 MLE doc asked for

Fit $\rho^*, \lambda^*, \gamma^*$ by maximum likelihood on draft outcomes Y_i.

That sounds like one joint fit. The subtlety is **what is held fixed vs re-simulated**.

Path A — Empirical NCAA rosters (fixed assignment)

If each row keeps **real** `(team_id, season)`:

- $\hat{A}_i$, teammates, L_C, and S_i are **fixed** for given λ, γ, θ.
- ρ **does not enter at all** — you did not re-run LG assign.
- MLE can estimate λ^*, γ^*, t^* (and maybe θ), **not** ρ^*, from this likelihood alone.

ρ **on this path** is matched separately — e.g. pick ρ so sim H_{sort} matches empirical H_{sort} (Alex wants H_{sort} in the paper). That is **calibration**, not draft MLE.

Path B — Full generative MLE (re-sim ASSIGN at each ρ)

For ρ^* **from draft data**, ρ must **move** the fitted object:

1. Fix ρ, run LG assign (seed choice matters).
2. Build L_C, S_i, SELECT (Gibbs + K draws or Bernoulli).
3. Evaluate likelihood of observed Y_i.
4. Optimize over ρ, λ, γ, t.

Hard parts:

Issue	Why it hurts
Stochastic ASSIGN	Same ρ → different rosters each seed → likelihood **noisy** unless you average over seeds
High dimensional	Full assign likelihood is not a simple product of softmax terms
Identifiability	ρ shapes L_C and H_{sort}; λ also moves SELECT — **tradeoffs**
Empirical $\hat{A}$ fixed	LG reshuffles seating only; ρ^* = "best **generative** homophily for draft," not "estimate NCAA's true matching process"

Path C — Two-step (fits this project so far)

Step	Estimate / match	Data
1. ASSIGN	ρ^* (or ρ band) via H_{sort}, coverage, overlap	Roster geometry — **not** the draft equation
2. SCORE + SELECT	λ^*, γ^*, t^* via MLE (after U gate)	Draft Y_i given rosters (empirical or sim at fixed ρ)

Alex PD15: fit **parameters to statistics of raw data**, not "make sim curve match hero bin-by-bin." H_{sort} is a natural ρ statistic; draft MLE is natural for λ, t (and maybe γ).

ρ^* is still a real goal — but likely **"ρ that matches sorting diagnostics"**, not **"ρ from the same draft likelihood as λ"** unless you commit to Path B.

Diagram (one glance)

```
ρ (ASSIGN)  →  rosters → L_C → S_i  (λ, γ, θ)
                                ↓
                 SELECT (t, K draws?)
                                ↓
                      Y_i draft
```

- **λ^*, t^*:** "Given rosters, how does congestion in **score** + Gibbs **select** relate to draft?"
- **ρ^*:** "What assign rule gives **realistic sorting** (H_{sort}, overlap)?" — **different data moment** unless you do full joint generative MLE.

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3. Still-open checklist

Question	Status	For t sweep now?
K draws vs literal softmax Bernoulli	**Open** for MLE; **K draws OK for sim**	Use K draws; confirm with Alex before MLE
ρ^* from draft likelihood vs H_{sort} match	**Open**	Fix ρ at PD17 value (e.g. 0.5); sweep t only
θ fixed from $F^{-1}(1 - K/N)$ vs estimated	Mostly fixed	Keep CDF rule for sweep
Joint ρ, λ, γ, t MLE	**Parked** after inverted-U gate	After PD20 success criterion

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4. One-liners for Alex

K draws:

> *"The softmax formula sums to 1 over the pool; MBB has K winners. For sim we use Gibbs weights and draw K without replacement so we still get K picks and can compare to PD17. For MLE we need to agree whether the likelihood uses that K-draw process or independent Bernoullis from softmax."*

ρ^*:

> *"ρ only moves draft predictions if we re-simulate rosters. On fixed empirical rosters, MLE fits λ and t, not ρ. ρ^* is either joint generative MLE (hard) or matching H_{sort}/overlap first, then fitting SELECT given that assign layer."*

Charles runs PDF when ready:

```
./scripts/convert_single_md_to_pdf.sh 3-Master_Plan/Alex_stuff/PD20_K_draws_and_rho_explainer.md
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